



Fig. 4: Security warning message

in Fig. 4. Click 'Yes' to continue. You will see a red chili moving from top to bottom of the screen. It means the program is working well. The captured file

with filename 'Track Saved' is saved on your hard disk in the same location as PTS.htm. This saved file can be run directly anytime using Internet Explorer without PTS.htm.

4. Now you can run the PTS.htm file again by repeating the above steps, but this time make multiple movements of the mouse cursor; from top to bottom, right to left and circular. You will see

Graphics Object Counter Simulation Using MATLAB

NISHANT MITTAL

Image processing has a significant role in scientific, industrial, space and security applications. It is increasingly finding applications in many other fields as well.

Digital image processing generally requires extensive experimental work involving software simulation and testing with large sets of sample images. MATLAB, C and CUDA are some popular computing tools used by scientists, students and engineers for graphics applications.

Presented here is a MATLAB program to count the number of graphics images in an image file. MATLAB has been used because of its dynamic capabilities and increasing usage in the industry. This program can detect even a 1mm diameter graphic object as explained below.

How images are identified and counted by the program?

In image processing, feature extraction and pattern recognition are implemented. Feature extraction is a special form of dimensionality reduction. Transforming the input data into a set of features is called feature extraction.

If the features extracted are carefully chosen, it is expected that the features set will extract the relevant information from the input data in order to perform the desired task.

In this project edge detection has been used to decide the number of objects. Edge detection is a fundamental tool in image processing, machine vision and computer vision.

There are many methods for edge detection, most of which can be grouped into two categories: search-based and zero-crossing based. Search-based methods detect edges by first computing a measure of edge strength (usually a first-order derivative expression such as the gradient magnitude) and then searching for local directional maxima of the gradient magnitude using a computed estimate of the local orientation of the edge (usually the gradient direction). The program can be made to detect objects of various sizes and shapes.

MATLAB program

MATLAB provides a very elegant graphical user interface (GUI) development platform for various applications. GUI allows users to perform tasks interactively through controls such as buttons and sliders. Within MATLAB,

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

that red chili follows the same path on the screen as the mouse cursor. ●

The author is a B.Tech in electronics and communications engineering who likes coding in Javascript. He has a keen interest in space technology (satellite communication) and mobile communication



GUI tools enable you to perform tasks like creating and customising plots, fitting curves and surfaces, and analysing and filtering the signals.

The plotter.m code of this project uses GUI tools to count the image objects. An image can be represented as the combination of thousands of tiny micro points called pixels, each representing particular RGB values. In MATLAB, an image may be defined as a two-dimensional function $f(x,y)$, where 'x' and 'y' are spatial plane coordinates. The amplitude of 'f' at any pair of coordinates (x,y) is called intensity of the image at that point. The term 'gray level' is often used to refer to the intensity of monochrome images. Colour images are formed by combination of individual 2D images.

The coordinate system representation of an image is given below in a matrix form:

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

Matrices in MATLAB are stored as